Assignment 3 — MVP Implementation & Demo

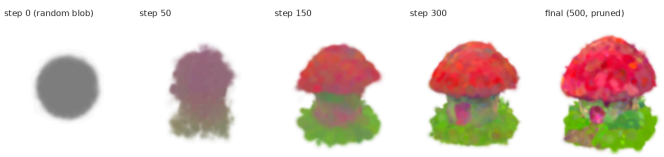
Minh Quan Dao (qd66) · Interactive Diffusion-Based 3D Generation Studio · Pillars: **Procedural Generation + Rendering**
Source: https://github.com/quandao10/diffusion-3d-studio · Web demo: https://quandao10.github.io/diffusion-3d-studio/ · Video: TODO – add after upload

1. Plan vs. Reality

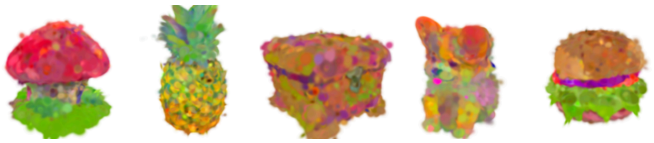
A2 MVP: one prompt → SDS from a frozen 2D diffusion model → optimise a **NeRF** → render novel views in an interactive viewer. **Built:** the same diffusion-guided loop running for real on an NVIDIA A100, but the 3D representation is **explicit 3D Gaussians** optimised with the official DreamGaussian code (commit `69a3c70`), not a NeRF. Reason: a from-scratch SDS-NeRF needs long optimisation and much more debugging for a one-person project; DreamGaussian's rasterizer makes a 500-step run take ≈47 s, so I could iterate and build the studio, monitor and stress test around it. Upstream code is used unmodified (one build-fix include); my own work is the instrumentation wrapper, the WebGL2 splatting viewer, the studio server/UI, the benchmark and the capture pipeline.

A2 feature (MoSCoW)	Status	Evidence
MUST text prompt; frozen diffusion prior; random cameras; SDS optimisation	**done**	SD 2.1-base fp16, CFG 100, annealed t 980→20; `results/*/training.log`
MUST NeRF + differentiable volume rendering	**changed**	3D Gaussians + differentiable CUDA rasterizer (DreamGaussian)
A2 “midterm proof”: non-zero SDS gradients reach the 3D parameters	**done**	per-step gradient norms of position/colour/opacity/scale/rotation all > 0 for all 500 steps (`metrics.jsonl`)
MUST interactive novel-view viewer	**done**	own WebGL2 splat viewer (orbit/zoom, splat-scale), CUDA turntables
SHOULD progress visualisation; seed/guidance/steps controls; save views	**done**	live loss/gradient/#Gaussian charts, snapshot scrubber, 8 views + MP4 per run
COULD 3DGS; mesh extraction; several variations	**done**	3DGS is now core; textured OBJ via upstream export; 5 prompts
COULD VSD comparison · stage-2 mesh refinement	**cut**	not implemented / not run

prompt (500 SDS steps, CFG 100, seed 0)	SDS time	total incl. mesh	final #Gaussians	peak VRAM
a small red mushroom house	46.5 s	123 s	311,695	4.35 GB
a ripe pineapple	46.3 s	103 s	293,427	4.81 GB
a wooden treasure chest	46.9 s	142 s	298,342	4.33 GB
a cute corgi puppy	46.7 s	124 s	312,254	4.34 GB
a delicious hamburger	45.7 s	122 s	281,563	4.31 GB



CUDA-rasterizer snapshots during SDS (“a small red mushroom house”): random 5k-Gaussian blob → steps 50/150/300 → final (312k Gaussians).



Novel views (45°) of the five generated objects (the hamburger was generated live in the video). Visible artefacts: soft splats, speckled over-saturated colour (CFG 100), multi-face (“Janus”) corgi.

2. Stress test (measured)

CUDA rasterizer, forward pass along a fixed 360° orbit, 120 frames after warm-up, synchronised timing, peak VRAM from PyTorch, dedicated A100 (`scripts/benchmark_renderer.py` → `evidence/performance.csv`).

Count axis = each Gaussian cloned k× with jitter, at 512².

resolution (311,695 Gaussians)	FPS	peak VRAM
256²	314.7	0.14 GB
512²	368.2	0.22 GB
1024²	175.1	0.50 GB
2048²	57.8	1.50 GB
4096²	16.8	5.30 GB

#Gaussians (512²)	FPS	peak VRAM
311,695	354.5	0.22 GB
1,246,780	144.6	0.78 GB
4,987,120	44.8	3.02 GB
19,948,480	11.7	11.97 GB
39,896,960	5.5	23.90 GB
79,793,920	**CUDA OOM (failure point)**	

The A2 viewer target (≥30 FPS) holds up to 2048² and ≈5 M Gaussians; the hard limit is memory (OOM at ≈80 M Gaussians). The browser viewer has its own ×1–×16 sweep in the studio, but on the server it ran on a GPU shared with another user's job, so I treat it as indicative only.

3. AI Utility

- **Runtime AI (diffusion prior):** works as planned: frozen SD 2.1-base supplies every gradient; no training of the diffusion model. The original `stabilityai` repo now returns HTTP 401, so I used a mirror of the same weights (documented).
- **Coding agent (Claude Code), listed in my tech spec:** very useful for environment work and scaffolding. From the logs, the env build started 11:44 and the first successful 500-step run finished 12:04. It found and fixed: `simple-knn` missing `<cfloat>` (gcc 13); CUDA runtime headers only under `targets/` on this server (`CPATH`); peak VRAM of 37 GB that turned out to be `cudnn.benchmark` switched on by the GUI seeding helper (fixed → 4.4 GB, the aborted log is kept); headless Chrome losing the WebGL context on Vulkan (switched to GL-EGL). It also wrote most of the wrapper, viewer, studio, benchmark and capture code.
- **Where it cost time:** every result still had to be checked by looking at renders and logs. A blank viewer first looked like a shader bug but was a lost GPU context, and the browser landing on a busy GPU could not be fixed. Time savings are not measured against a baseline. The video voiceover is synthetic (Kokoro-82M TTS), and the on-screen captions match it.

4. Roadmap to Final

- **Geometry/consistency:** try upstream MVDream/ImageDream multi-view guidance to reduce Janus faces; tune CFG/timestep schedule and negative prompts; run stage-2 mesh refinement (`main2.py`).
- **Studio:** side-by-side seed/guidance comparison, run history with parameters, mesh/Gaussian toggle, websocket progress instead of polling.
- **Viewer performance:** GPU sorting and level-of-detail for multi-million-Gaussian scenes; measure the browser viewer on a dedicated GPU.
- **Stretch (only after the above):** VSD comparison, as described in A2.

5. Build / run

Linux + NVIDIA GPU (≥8 GB). `bash scripts/setup_env.sh` → `conda activate dreamgaussian` → `GPU=0 bash scripts/run_text_to_3d.sh "a small red mushroom house" mushroom_house` → `GPU=0 python server.py` and open `http://127.0.0.1:8549`. Full instructions, versions and evidence are in the repository README and `evidence/`. Demo video: 179.6 s, real-time screen capture of the studio (two waits fast-forwarded 8× and labelled on screen).